

CS661: Project Proposal Report

MarketXray: High-Frequency Market Microstructure Analytics

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1 Introduction

Modern financial markets operate at sub-millisecond speeds, generating millions of data points per minute. The Limit Order Book (LOB) is the core engine of this ecosystem, containing every bid, ask, and cancellation across multiple price levels. Standard visualizations fail to capture the transient dynamics of market microstructure, such as algorithmic spoofing, latent liquidity, and high-frequency arbitrage opportunities.

Our project aims to develop **MarketXray**, a high-performance, interactive visual analytics engine for market microstructure. By leveraging tick-level order book data, we will build a web-based system that allows users to peer into the "plumbing" of the stock market. The system will decode the raw data to tell a compelling story about institutional trading behavior, market impact, and liquidity flash crashes.

2 Data Source

The primary data source will be high-frequency L2 Tick Data (NASDAQ ITCH data / Binance L2 Order Book snapshots). This dataset contains:

- Timestamped additions, executions, and cancellations of limit orders.
- Deep order book snapshots (up to 50 levels of bids and asks).
- Trade executions and order flow imbalances.

3 Specific Tasks

To handle the extreme scale of tick data, we will perform the following tasks:

- **High-Speed Ingestion (Rust):** Parse the raw, binary-level tick data asynchronously into a structured format to ensure maximum throughput.
- **Microstructure Analytics (C++/Rust):** Compute complex derived metrics in real-time, such as Order Flow Imbalance (OFI), Volume Weighted Average Price (VWAP), and market impact models.
- **Data Visualization:** Build interactive frontend representations of LOB dynamics using D3.js and WebGL.
- **Interactive Querying:** Allow users to isolate specific micro-events, filter by trade volume, and zoom into millisecond-level timeframes.

4 Visualization Tasks

We will implement the following interactive visual analytics plots:

4.1 3D Limit Order Book Depth Surface

Standard 2D depth charts hide historical context. We will render a 3D surface plot where the X-axis is time, the Y-axis is price level, and the Z-axis (and color heatmap) represents liquidity volume. Users can rotate the plot and zoom into specific timestamps to visualize "liquidity walls" forming and disappearing.

4.2 Algorithmic Spoofing & Layering Detector

This visualization tracks the lifecycle of canceled orders. A scatter plot will map order size against time-to-cancellation. Interactive brushing will allow users to highlight clusters of large, rapidly canceled orders—a key indicator of market manipulation (spoofing).

4.3 Order Flow Imbalance (OFI) vs. Price Action

A dual-axis interactive line chart correlating the net difference between aggressive buy and sell orders (OFI) with the underlying asset's mid-price. Users can adjust the lookback window to see how leading OFI indicators predict short-term price movements.

4.4 Arbitrage Opportunity Network Graph

A dynamic network graph (nodes = assets/exchanges, edges = exchange rates/spreads) that visualizes transient triangular arbitrage opportunities. Edges will briefly flash red when market inefficiencies occur, allowing users to pause the timeline and inspect the spread calculation.

4.5 Market Impact & Slippage Simulator

An interactive "what-if" tool. Users input a hypothetical large market order size. The visualization will instantly calculate and display the expected slippage by "walking the book" against the historical LOB data at that exact millisecond, rendering a bar chart of filled orders across different price levels.

5 Overall Solution

MarketXray will be an integrated web application. Because plotting millions of data points on a browser is a bottleneck, the heavy mathematical aggregations and microstructure modeling will be handled by our C++ backend, while the Rust layer efficiently pipelines data via WebSockets to a React/D3.js frontend. The interface will feature synchronized zooming across all plots, allowing analysts to cross-reference a price spike with spoofing behavior and order flow imbalances simultaneously.

6 Tech Stack

- **Data Ingestion & Pipeline:** Rust (Tokio)
- **Analytics Engine:** C++ (Boost, Eigen for quantitative aggregations)

- **Frontend:** JavaScript/TypeScript (React.js), D3.js, WebGL
- **Communication:** WebSockets / WebAssembly (WASM)

7 Team Members & Responsibilities

Backend (Data Ingestion Pipeline in Rust, Analytics Engine in C++)

- Ashish Gautam (240210)
- Chanakya Navghare (240293)
- Kartik Agarwal (240526)

Frontend (Visualisation using React.js, D3.js, and WebGL)

- Aadi Chhajed (240004)
- Chhaya (240307)
- Karan Chauhan (240519)
- Manthan Vinay Khetade (240631)
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